Polyphase P4 Code Usage for Target Detection of Pulse Radar Using Matlab and GNU Radio

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***Abstract*—This paper presents the investigation of polyphase P4 code usage in pulse radar for target detection as part of range resolution. As one of essential parameters in pulse radar, the range resolution is acquired to determine the relative range between two or more close targets in order to be detected as different targets. The generation of phase-coded pulse signal and its processing are performed in Matlab® and GNU Radio by utilizing audio waves. From the result, it shows that the phase-coding with polyphase P4 code could provide good compression to the pulse signal. In addition, by making the code longer it generates side lobe level less than –40dB for the code with length of 2000.**

Keywords—Audio waves; GNU radio; polyphase P4 code; pulse radar; target detection.

I. INTRODUCTION

Radio detection and ranging, known as radar, is an object detection system that utilizes electromagnetic waves to determine range, altitude, and velocity of an object [1]–[2]. In principle, it works by transmitting electromagnetic waves to a certain space in which some of the waves will be reflected back by the object in that space. The reflected wave is then captured and processed so that the information about the object, such as its range, velocity, etc. can be gained. Basically, radar can be classified into two based on its waveform, namely pulse radar and continuous wave (CW) radar [2]. In pulse radar, it radiates a train of pulse signals periodically where the radiated pulse signals are usually in modulated form. On the other hand, CW radar radiates signal continuously. Hence, CW radar can also be seen as pulse radar with 100% duty cycle.

In pulse radar, the range resolution is determined by the pulse width [3]. A good range resolution can be achieved by narrowing the pulse width. By narrowing the pulse width, the average transmitted power will decrease. The decrement in average power will cause the transmitted signals have no significant power to reach the desired range. To compensate for this drawback, the individual pulse signal power should be increased. Moreover, the problem of range resolution can be solved by modulating the pulse signal which is meant to widen the bandwidth of the transmitted signal [4]. Since the range resolution is related to the signal bandwidth inversely, therefore the modulation which improves range resolution without narrowing the pulse width become possible.

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There are a few common modulation schemes used in radar, namely linear frequency modulation (LFM) [5]–[6], stepped frequency modulation (SFM) [7]–[8], and phase-coded pulse [9]. Phase-coded pulse is basically meant to modulate a carrier signal with a certain code in which the code is saved in the phase of signal. This process can also be imagined as dividing the pulse into N sub-pulses, where N is the length of code. The the length of sub-pulses will define the range resolution. In general, the phase-coded pulse can be grouped in two categories, i.e. biphase and polyphase.

In this paper, the usage of polyphase P4 code in pulse radar for target detection as part of range resolution is presented. The processes including the generation of phase-coded pulse signals through the processing of received signals are simulated in Matlab® and GNU Radio. The result of radar range resolution will show the detection of multiple targets. From the range profile, it will be seen how close the relative range between two or more close targets in order to be detected as different targets.

II. POLYPHASE P4 CODE AND CORRELATION PROCESSING

A. Brief Overview of Polyphase P4 Code

Phase-coded pulse consists of N sub-pulses where the phase is intentionally carried from sub-pulse to sub-pulse to achieve the desired main lobe and side lobe responses. In general, the length of an individual sub-pulse in pulse radar defines the range resolution. In pulse radar, there are two categories of phase-coded pulse, i.e. biphase and polyphase. One of important parameters of phase-coded pulse is side lobe level. It is already known that the side lobe level of phase-coded pulse decreases along with the increase of the code length [10], which is N.

Basically, polyphase P4 code is a phase code with the value of phase code expressed in (1) [11].

$$\phi_m = \frac{2\pi}{N}(m-1)\left(\frac{m-1-N}{2}\right) \tag{1}$$

with $1 \leq m \leq N$. The code element is defined as (2).

$$s_m = e^{j\phi_m} \tag{2}$$

It shows from (2) that the generated code sequence is in a form of complex number.

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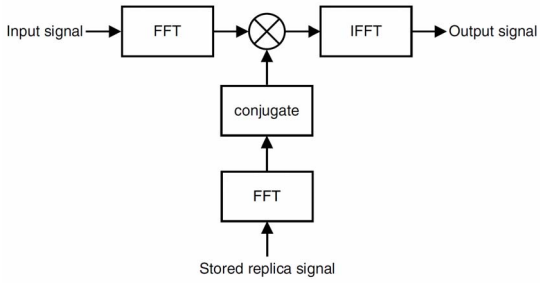


Fig. 1. Implementation of correlation processing using FFT technique.

B. Correlation Processing

The range resolution in radar pulse as one of essential parameters can be improved by using very short pulses. Unfortunately, short pulses decrease the average transmitted power which can hinder the operation of radar. To achieve good range resolution while maintaining the average transmitted power, pulse compression technique is one of solutions to be used. There are two kinds of pulse compression technique, i.e. analog and digital [11]. Here, the compression technique for analog pulse known as correlation processing is discussed. In this work, the correlation processing is used for detecting object in the phase-coded pulsed radar.

In correlation processing, pulse compression is accomplished by adding exponential (frequency or phase) modulation to a long pulse at transmission and a matched filter in the receiver to compress the received signal. Since matched filter is a linear time invariant system, therefore the output of matched filter can be described as convolution between its input and impulse response. In time domain, this relation is expressed in (3).

$$s_o(t) = \int_{-\infty}^{\infty} s_t(\tau) s^*(t-\tau) dt \tag{3}$$

Meanwhile, in frequency domain (3) can be rewritten as (4).

$$S_o(f) = S_t(f) S_t^*(F) \tag{4}$$

It shows that (3) is simple convolution properties of Fourier transform. Figure 1 shows the implementation of correlation processing using fast Fourier transform (FFT) technique. When using pulse compression, it is desirable to use modulation schemes which is available to accomplish a maximum pulse compression ratio. Furthermore, the modulation schemes is also useful to suppress significantly the side lobe level of compressed waveform. In actual, the weighting function (windows) in frequency domain can be applied to reduce the side lobe level. However, this approach will cost in the loss of main lobe resolution.

III. Simulation and Discussion

Simulation of target detection for polyphase P4 code in pulse radar is performed in Matlab® and GNU Radio. Figure 2 shows the diagram block of basic process performed in the simulation using Matlab®. The system involves a

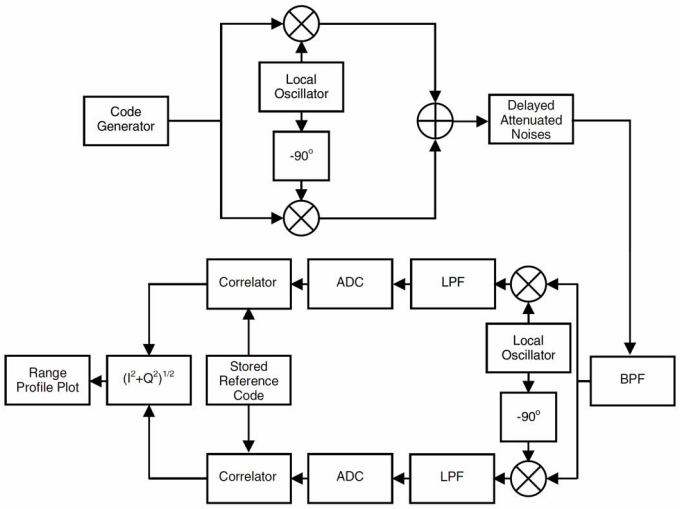


Fig. 2. Diagram block of basic process performed in simulation.

TABLE I. Specification of Pulse Radar.

Paramter	Value
Sampling frequency	10MHz
Carrier frequency	2.5MHz
Speed of light	3×10^8 ms^{-1}
Length of code (N)	2000
Duty cycle of pulse	50%
Pulse repetition frequency	500

quadrature modulator in transmitter and another quadrature demodulator in receiver. The generated code is a complex number which is separated in real and imaginary part, and used to modulate the carrier in the quadrature modulator. The modulated signal is then delayed, attenuated, and added with some random noises. This signal is then input into the receiver which will be processed in two different channels, i.e. in-phase and quadrature-phase. In each channel, there are ADC and matched filtering processes. The signals from two channels are then combined. At the end of receiver, a range profile plot can be obtained in which it shows the range of target. Table I summarizes the specification of pulse radar used in the simulation. Based on the specification, the maximum unambiguous range and the range resolution can be determined, i.e. are 150km and 75m, respectively.

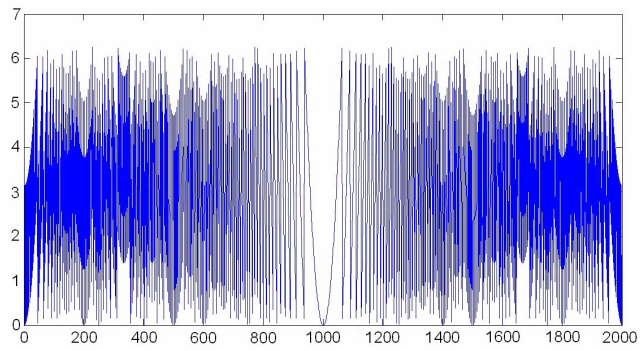


Fig. 3. Phase code with code length of 2000.

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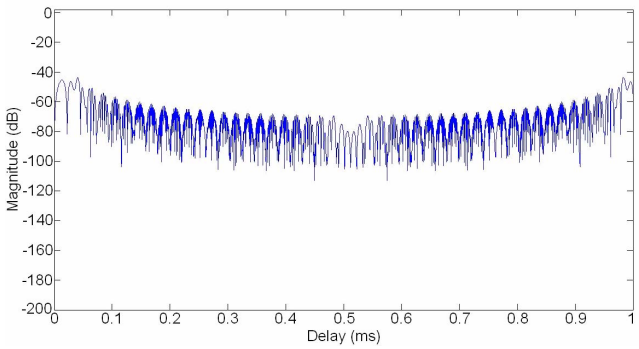


Fig. 4. Autocorrelation of code sequence with code length of 2000.

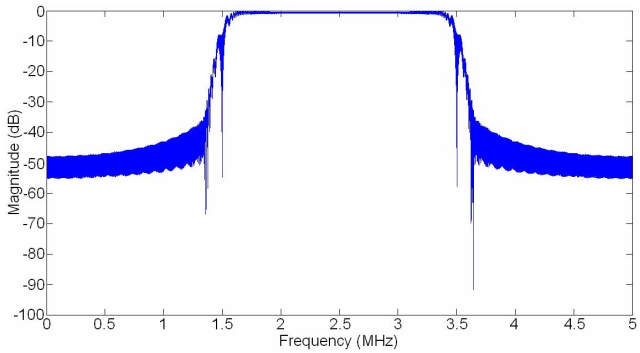


Fig. 5. Frequency spectrum of modulated signal.

Figure 3 shows the generated phase code with code length of 2000 based on (1). Meanwhile, the autocorrelation of code sequence is plotted in Fig. 4. It can be seen that the side lobe level is very low, i.e. less than –40dB. The frequency spectrum of modulated signal is depicted in Fig. 5. It shows the effective bandwidth is obtained if the pulse signal is around 2MHz with the center frequency of 2.5MHz in which it is relevant to the calculation result.

Figures 6 and 7 show the radar range profile produced by the target detection process from the delayed signals. Each signal is assumed come from a different object with a certain value of radar cross section (RCS). In Fig. 6, it shows five different impulses in which each impulses represents a single target. The RCS and delay for the targets are tabulated in Table II. In this case, five different targets are detected as an individual object. While in Fig. 7, the targets with the RCS and delay summarized in Table III are set to be closed each other, in which target#1 and target#2 differ 15 samples in delay, target#2 and target#3 differ 5 samples in delay, and target#4 and target#5 differ 25 samples in delay. If the differences in delay is converted into the range, the distances between target#1 and target#2, target#2 and target#3, target#4 and target#5 are 225m, 75m, 375m respectively.

For target#2 and target#3 with 5 samples difference in delay, or equal to 75m, these two targets cannot be detected as different target due to their relative ranges to each other are equal to the range resolution of the radar. In case for target#1 and target#2 (or target#4 and target#5), their relative ranges to each other are greater than the radar range resolution. Therefore, they should can be detected as different targets

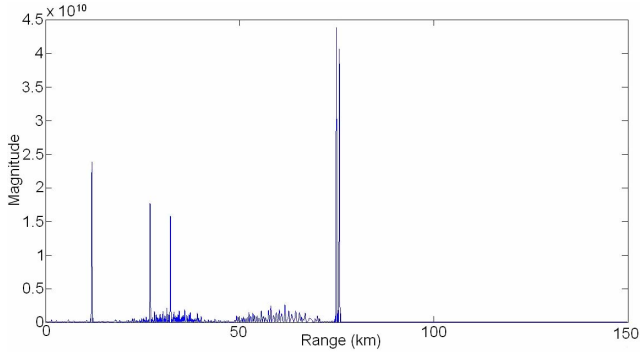


Fig. 6. Radar range profile with 5 different impulses.

TABLE II. RCS AND DELAY FOR FIVE DIFFERENT IMPULSES.

Target Number	RCS(m^2)	Delay (sample)
Target#1	1	750
Target#2	1	1750
Target#3	1	2100
Target#4	5	4950
Target#5	5	5000

theoretically. However, it is seen that the targets cannot be differed by the pulse radar. A few possible reasons are errors in the processing and the side lobe of matched filter output. Errors in the processing such as rounding and the lack of number of samples are possible to happen as this simulation was carried out digitally. The effect of side lobe level is possible to be taken into account. Although its level is about –40dB below the main lobe, it can produce error in detection especially for a very close target.

Furthermore, the simulation of phase-coded pulse radar for target detection in GNU Radio was conducted by using the help of GNU Radio Companion application. It should be noted

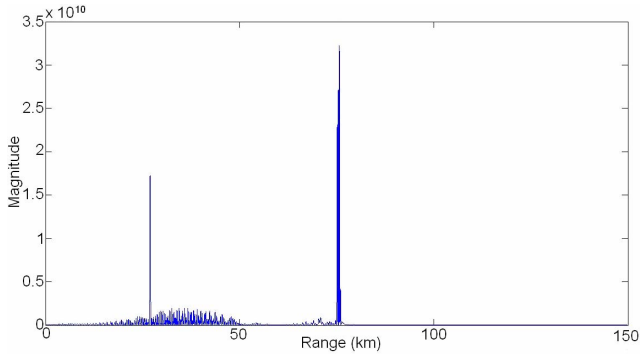


Fig. 7. Radar range profile with 5 different targets and closer distance.

TABLE III. RCS AND DELAY FOR FIVE DIFFERENT TARGETS AND CLOSER DISTANCE.

Target Number	RCS(m^2)	Delay (sample)
Target#1	1	1735
Target#2	1	1750
Target#3	1	1755
Target#4	5	4975
Target#5	5	5000

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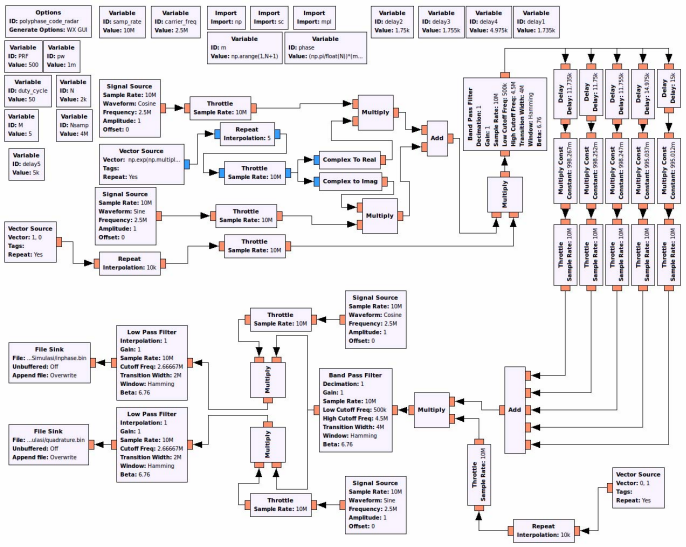


Fig. 8. Block diagram of simulation in GNU Radio Companion application.

that the simulation in GNU Radio is carried out by utilizing audio waves. Figure 8 shows the block diagram of simulation used in GNU Radio Companion application. Basically, the block diagram shown in Fig. 8 has similar concept with the one visualized in Fig. 2 which consists of transmitter and receiver. In this simulation, the received signals, after being processed by the receiver, are stored to a binary file "inphase.bin" and "quadrature.bin". These files are processed by using a Python script that uses some Python libraries such as numpy, scipy, and matplotlib.

Figure 9 shows the radar range profile produces by GNU Radio with the RCS and delay summarized in Table II. It is seen that the result is similar to the one in Fig. 6. In case the target is set close each others as shown in Fig. 10 and tabulated in Table III, it is only two dominant impulses shown in the range profile plot. It can be seen that the impulse also consists of a few sub-impulses, but these sub-impulses are not clearly separated in order to determine which one which represents target#1, target#2, or target#3. For the second impulse which consist of two targets, it shows two dominant impulses which may represent target#4 and target#5.

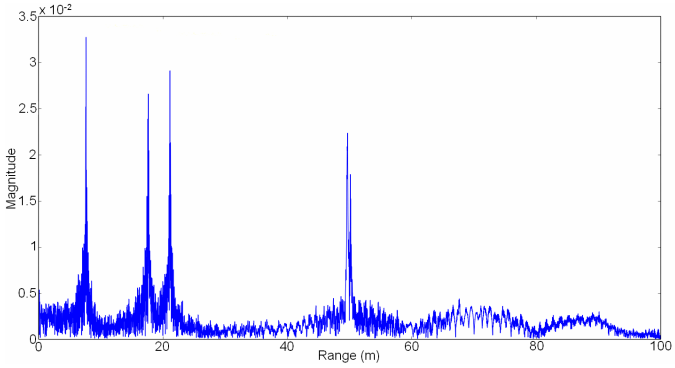


Fig. 9. Radar range profile with 5 different impulses using GNU Radio.

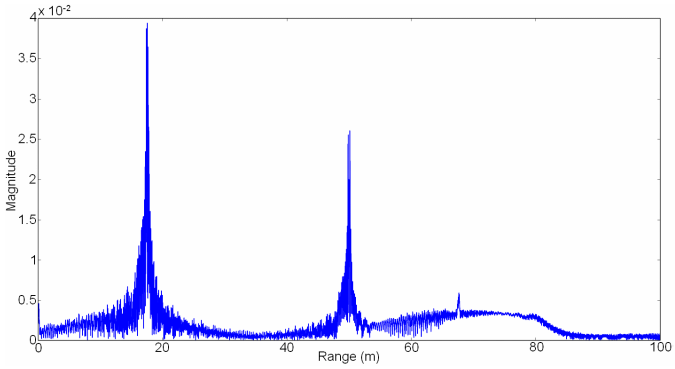


Fig. 10. Radar range profile with 5 different targets and closer distance.

IV. CONCLUSION

The investigation of target detection for pulse radar using phase-coded with polyphase P4 code has been demonstrated using Matlab® and GNU Radio. The simulation in Matlab® has been performed based on pulse radar specification, meanwhile for GNU Radio, the simulation is carried out by utilizing audio waves. From the simulation, it has been shown that results show a quite good result, both in Matlab® and GNU Radio. The radar range profile has produced a good approximation to the range of multiple targets. Therefore, it can be inferred that the usage of polyphase P4 code improved the radar range resolution.

REFERENCES

[1] M. Skolnik, *Introduction to Radar Systems*, 3rd ed., McGrawHill, 2002.

[2] M. Skolnik, *Radar Handbook*, 3rd ed., McGrawHill, 2008.

[3] M. A. Richard, J. A. Scheer, and W. A. Holm, *Principles of Modern Radar: Basic Principles, SciTech Publishing*, 2010.

[4] B-C. Wang, *Digital Signal Processing Techniques and Applications in Radar Image Processing*, 1st ed., Wiley-Interscience, 2010.

[5] E. J. Amin, A. B. Suksmono, and A. Munir, "Accuracy analysis of FM chirp in GNU radio-based FMCW radar for multiple target detection," in *Proceeding of International Conference on Computer, Control, Informatics and Its Applications (IC3INA)*, Bandung, Indonesia, Oct. 2014, pp. 115–119.

[6] I. Intyas, R. Hasanah, M. R. Hidayat, B. Hasanah, A. B. Suskmono, and A. Munir, "Improvement of radar performance using LFM pulse compression technique," in *Proceeding of International Conference on Electrical Engineering and Informatics (ICEEI)*, Denpasar, Indonesia, Aug. 2015, pp. 302-307.

[7] Azizah, A. B. Suksmono and A. Munir, "Signal processing of range detection for SFCW radars using Matlab and GNU radio," in *Proceeding of International Conference on Computer, Control, Informatics and Its Applications (IC3INA)*, Bandung, Indonesia, Oct. 2014, pp. 145–148.

[8] S. Aulia, A. B. Suksmono, and A. Munir, "Stationary and moving targets detection on FMCW radar using GNU radio-based software defined radio," in *Proceeding of International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Denpasar, Indonesia, Nov. 2015, pp. 468-473.

[9] S. Plata and M. Pergol, "On orthogonality of Linear and Non-Linear frequency modulated pulses," in *Proceeding of 15th International Radar Symposium (IRS)*, Gdansk, Poland, Jun. 2014, pp. 1-6.

[10] N. Levanon and E. Mozeson, *Radar Signals*, Wiley-IEEE Press, 2004.

[11] B. R. Mahafza, *Radar Systems Analysis and Design Using MATLAB*, 2nd ed., Chapman & Hall/CRC, 2000.

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